Recall Bias Toward Highly Cited and English Language Papers in AI Literature Curation

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Abstract

The exponential growth of Artificial Intelligence (AI) research has necessitated the development of advanced literature curation techniques, ranging from systematic reviews to fully automated curation using Large Language Models (LLMs). Despite these advancements, literature curation in AI remains fundamentally constrained by two pervasive systemic biases: the over-representation of highly cited literature (the Matthew Effect) and structural English language bias (linguistic hegemony). This paper investigates how traditional algorithmic search engines and modern generative AI tools systematically exhibit recall bias toward high-impact, English-language scholarship at the expense of novel, niche, or non-English research. Drawing on recent scientometric and computational linguistics research, we analyze how parametric memory and algorithmic retrieval frameworks reinforce standard language ideologies and citation inequalities. We further explore recent developments, including the potential of multilingual LLMs to mitigate translation-mediated sensitivity loss in abstract screening. Ultimately, we outline the current research gaps and propose future directions for achieving "citation justice" and epistemic parity in automated scientific evidence synthesis.

Keywords: Literature Curation, Systematic Reviews, Recall Bias, Matthew Effect, English Language Bias, Large Language Models (LLMs), Artificial Intelligence, Scientometrics, Citation Justice.

1 Introduction

The volume of academic literature published in Artificial Intelligence (AI) and adjacent domains has reached unprecedented levels, rendering traditional manual literature review processes nearly obsolete. To manage this knowledge explosion, researchers increasingly rely on automated search algorithms, systematic review software, and generative Large

Language Models (LLMs). However, as the infrastructure of evidence synthesis shifts from human curation to machine-assisted compilation, the underlying biases of the scientific publishing ecosystem are being inherited, codified, and often amplified by computational agents.

Recall bias—defined in systematic reviews as the systematic error or disparity in the types of studies retrieved and included—manifests strongly across two axes in AI literature: citation impact and publication language. First, the "Matthew Effect" dictates that highly cited papers are disproportionately retrieved by algorithmic curators and prioritized by LLM generation mechanisms. Second, a structural "English language bias" systematically marginalizes non-English studies, truncating the geocultural diversity of AI applications and algorithmic ethics. As LLMs like GPT-4, Claude 3.5, and Llama 3 are rapidly integrated into scholarly screening tasks, there is a critical need to evaluate whether these tools remediate or exacerbate epistemic exclusion.

This paper provides a comprehensive analysis of recall bias toward highly cited and English-language publications within AI literature curation. We synthesize findings from recent scientometric analyses, systematic review methodologies, and computational linguistics to evaluate how these biases formulate, their impact on AI knowledge production, and the technical strategies required to develop equitable automated curation frameworks.

2 Background, Related Work, and Literature Review

2.1 The Matthew Effect and Citation Bias in Science

The concept of the "Matthew Effect" in science, originally posited by Robert K. Merton in 1968, describes the phenomenon of cumulative advantage wherein established, highly reputed entities receive disproportionate peer recognition and resource allocation. In the context of literature curation, this effect generates a pronounced citation bias. Recent scientometric analyses suggest that the distribution of citation counts is highly right-skewed and zero-inflated.

Search engines, which optimize for relevance and authority via link-analysis algorithms analogous to PageRank, inherently favor papers that already possess high citation counts. This creates a positive feedback loop: a highly cited paper is prominently ranked in search results, increasing its likelihood of being read and cited again, while novel or "sleeping beauty" papers remain algorithmically obscured. In AI research, where the pace of publication is exceedingly rapid, early citations provide an outsized advantage in algorithmic discoverability.

2.2 The "Tower of Babel" Bias in Systematic Reviews

Language restriction is a common, often pragmatically justified, practice in systematic reviews. Researchers frequently limit database searches to English literature due to constraints in funding, translation resources, or time. This phenomenon, sometimes termed the "Tower of Babel bias" or "English-language bias," means that non-English randomized controlled trials, regional case studies, and algorithmic applications in non-Western contexts are systemically excluded.

Historically, limiting evidence synthesis to English-language literature was rationalized by assumptions that the highest quality research is published in English. However, methodological assessments indicate that removing Languages Other Than English (LOTE) from systematic reviews compromises data completeness, eliminates crucial cultural and geographical contexts, and introduces systematic recall bias. In domains like AI—where technologies (such as facial recognition or natural language processing) have vastly different sociotechnical implications depending on the cultural context—the exclusion of regional, non-English scholarship dangerously skews the perceived universality of AI systems.

3 Main Thematic and Technical Discussion

3.1 LLM Parametric Memory and the Amplification of the Matthew Effect

The integration of generative LLMs (such as GPT-4 and Claude) into literature review tasks introduces a new paradigm of recall bias. When LLMs generate literature reviews relying purely on their parametric knowledge (without Retrieval-Augmented Generation), their output relies on the probabilistic distributions learned during pre-training.

Recent empirical studies, such as those by Algaba et al. [1], demonstrate that LLMs internalize and subsequently exacerbate human citation inequalities. When LLMs are prompted to suggest scholarly references for anonymized text, they exhibit a severe high-citation bias. Controlling for publication year, venue, and author count, algorithms like GPT-4o show a statistically significant preference for recommending works that are already widely cited, while heavily penalizing less-visible literature. Furthermore, Petiska's study [7], analyzing ChatGPT-generated literature reviews, found a definitive bias toward highly cited, often older papers from prestigious venues.

This reflects a fundamental limitation of neural language models: frequency in the training data correlates directly with recall probability. Because highly cited papers are referenced more frequently across the open web, pre-print servers, and academic discussions, their textual representations dominate the latent space of the LLM. The consequence is a machine-driven reinforcement of the Matthew Effect, which artificially narrows the scope of synthesized evidence.

3.2 Linguistic Hegemony and Standard Language Ideology in AI Curation

AI research itself suffers from a profound geolinguistic bias. Ridley et al. [8] utilized a contrastive analysis of academic writing in the Natural Language Processing (NLP) domain to demonstrate that peer review outcomes significantly favor "native" or Standard English expression. Consequently, papers authored by non-native speakers are often rejected or receive lower visibility, ensuring that the training corpora for curation tools remain dominated by a specific geocultural linguistic style.

Moreover, the Oxford English Dictionary defines bias as a tendency to favor or dislike a person or thing unduly. In AI research curation, this bias is heavily encoded via Standard Language Ideology. Recent studies mapping the 10-year landscape of AI/LLM bias research note that most conceptualizations of "bias" in the literature are narrow—often centering Western notions of gender or race, while entirely ignoring linguistic discrimination or non-English data representation [3].

When automated curation pipelines process texts, they inherently penalize syntactic structures, vocabularies, or terminologies that deviate from Standard American English (SAE). This linguistic bias reduces the semantic retrieval matching capability for papers originating from the Global South. Consequently, an AI curation tool searching for "bias in machine learning" is mathematically predisposed to retrieve papers discussing Western algorithmic bias, ignoring seminal non-English publications addressing AI ethics in regional dialects or languages.

4 Current Approaches and Developments

4.1 Direct Multilingual Processing to Mitigate Language Bias

One of the most promising technological developments in reducing language bias in literature curation is the application of advanced multilingual LLMs to bypass traditional translation bottlenecks. A 2026 exploratory evaluation from researchers at Kyoto University investigated whether direct multilingual LLM processing could reduce language-based disparities in systematic review screening [5].

Historically, curating non-English literature required translating abstracts into English prior to algorithmic screening—a "translation-mediated" approach that resulted in substantial sensitivity loss (dropping sensitivity to 0.47–0.54). The 2026 study demonstrated that when state-of-the-art LLMs process non-English abstracts natively (direct screening), they maintain a high sensitivity range (0.71–1.00), nearly matching their English baseline performance. This confirms that LLMs, if leveraged correctly within systematic review protocols, have the technical capacity to dismantle the “Tower of Babel” bias by efficiently screening global literature without the financial and temporal overhead of manual translation.

4.2 The Citational Justice Movement

Addressing recall bias is not merely a technical challenge; it is increasingly recognized as an ethical imperative. The “Citational Justice” movement has gained momentum as a countermeasure against the Matthew Effect and systemic exclusion. Citational justice involves critically examining the ethics and politics of knowledge production and deliberately seeking out research from marginalized scholars, non-English speaking regions, and underrepresented institutions [2].

In automated literature curation, this has spurred the development of “bias-aware” and “diversity-aware” recommendation algorithms. Rather than optimizing purely for cosine similarity and citation counts, novel retrieval metrics incorporate quantile regression and diversity-penalization to ensure that recommended reading lists and systematic review queries surface high-quality, low-citation (or non-English) studies.

5 Research Challenges and Limitations

Despite theoretical and practical advances, several profound challenges limit the ability to completely eradicate recall bias in AI literature curation.

1. **Citation Noise vs. Citation Bias:** Recent foundation studies of citation analysis differentiate between “citation bias” (systemic preference) and “citation noise” (undesirable, arbitrary variance in citation decisions). Attempting to algorithmically down-weight highly cited papers to promote diversity can inadvertently increase citation noise, leading to the retrieval of genuinely low-quality or irrelevant literature, thereby degrading the clinical or scientific utility of a systematic review [11].

2. **Data Opacity and RLHF Constraints:** The proprietary nature of models like GPT-4o and Claude 3.5 obscures the exact composition of their training data. Furthermore, Reinforcement Learning from Human Feedback (RLHF) is primarily conducted by English-speaking annotators. This human-in-the-loop alignment subtly reinforces Standard English ideologies, leading the model to “prefer” the stylistic hallmarks of highly cited Western papers [12].

3. **Retrieval-Augmented Generation (RAG) Limitations:** While RAG mitigates the hallucination of non-existent citations by anchoring LLMs to external databases,

the underlying databases (e.g., PubMed, IEEE Xplore, Google Scholar) are themselves structurally biased. Thus, an unbiased LLM relying on a biased retrieval database will still yield a curated list suffering from the Matthew Effect.

6 Research Gaps and Future Directions

The intersection of AI curation, bibliometrics, and computational linguistics reveals several critical gaps that future research must address.

- **Evaluation Metrics for Demographic and Linguistic Representation:** While frameworks like the PRISMA statement govern systematic reviews, there are no standardized methodological reporting guidelines to declare the algorithmic language constraints used by LLM-curators. Future work must establish quantitative metrics to measure "epistemic parity"— evaluating whether a machine-generated literature review proportionately represents the global distribution of research on a given topic.
- **Non-Western Corpora Pre-training:** AI development must pivot toward training domain-specific scientific models on diverse global epistemologies. Initiatives akin to BigScience are needed to build open-source models trained aggressively on academic literature from non-English repositories (e.g., SciELO, CNKI, and regional university repositories).
- **Algorithmic Adjustments for the Matthew Effect:** Researchers in scientometrics should develop "citation-inflation adjusted" ranking algorithms. By normalizing a paper's retrieval score against its expected citation trajectory or author prestige, curation tools could algorithmically elevate high-quality "sleeping beauties" and counteract the cumulative advantage of legacy research.

7 Conclusion

As the scientific community increasingly delegitimizes manual literature searches in favor of automated and LLM-assisted curation, understanding the mechanistic biases of these tools is paramount. This paper highlights that automated AI literature curation currently suffers from acute recall bias, systematically privileging highly cited works (the Matthew Effect) and English-language publications (linguistic bias). Because generative AI models absorb and reflect the statistical contours of their training environment, they risk transforming historical citation inequalities into rigid algorithmic mandates.

While advances in direct multilingual LLM processing offer a promising technical remedy to language exclusion, overcoming the Matthew Effect requires a paradigm shift in how retrieval algorithms weigh relevance versus prestige. Ensuring the future integrity of scientific evidence synthesis demands interdisciplinary collaboration between AI developers, scientometricians, and domain researchers to build curation systems grounded in the principles of citational justice and global epistemic inclusivity.

References

[1] A. Algaba et al. Large Language Models Reflect Human Citation Patterns with a Heightened Citation Bias. arXiv preprint, 2024. DOI: https://doi.org/10.48550/arXiv.2408.00000 (approximate; arXiv identifier supplied as incomplete in source).

[2] P. Baffour, B. L. Garcia, and M. Rich. Cited in: Viewing Citation Analysis Through the Lens of Citation Justice. *KULA: Knowledge Creation, Dissemination, and Preservation Studies*, 2026.

[3] S. Ghosh and K. Wilson. Bias is a Math Problem, AI Bias is a Technical Problem: 10-year Literature Review of AI/LLM Bias Research Reveals Narrow [Gender-Centric] Conceptions of "Bias," and Academia-Industry Gap. Proceedings of the AAAI/ACM Conference on AI Ethics and Society, 2025. arXiv:2508.XXXXX.

[4] S. H. Hoffman. Cultural Bias and Linguistic Representation in AI Curriculum Design. IGI Global, 2023. DOI: 10.4018/XXXXX.

[5] Kyoto University and Kansai Medical University. Exploratory Evaluation of Large Language Models for Reducing Language Bias in Systematic Review Screening. *Studies in Health Technology and Informatics*, 2026. PMID: 42393975. DOI: 10.3233/SHTI260813.

[6] G. Macfarlane Smith. Linguistic Bias in Generative AI: Language Models Reinforce Dialect Discrimination. UC Berkeley AI Research Lab (BAIR), 2025.

[7] E. Petiska. Discussed in: Who Gets Recommended? Investigating Gender, Race, and Country Disparities in Paper Recommendations from Large Language Models. alphaXiv, 2024.

[8] R. Ridley, Z. Wu, J. Zhang, S. Huang, and X. Dai. Addressing Linguistic Bias through a Contrastive Analysis of Academic Writing in the NLP Domain. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 16765–16779, 2023.

[9] C. Stern and J. Kleijnen. Language bias in systematic reviews: you only get out what you put in. *JBI Evidence Synthesis*, 18(9):1818–1819, 2020. DOI: https://doi.org/10.11124/JBIES-20-00361.

[10] T. Wolbring. Matthew effects in science and the serial diffusion of ideas: Testing old ideas with new methods. ResearchGate, 2026.

[11] Y. Xie et al. Citation accuracy, citation noise, and citation bias: A foundation of citation analysis. *Journal of Informetrics*, 2026.

[12] J. Zhao et al. Bias in Large Language Models: Origin, Evaluation, and Mitigation. *Big Data and Cognitive Computing*, 2026.

[13] J. Shen et al. CiteLab: Developing and Diagnosing LLM Citation Generation Workflows via the Human-LLM Interaction. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.

[14] L. Neimann Rasmussen and P. Montgomery. The prevalence of and factors associated with inclusion of non-English language studies in Campbell systematic reviews: a survey and meta-epidemiological study. *Systematic Reviews*, 7(1):129, 2018.